

HEARTLAND Protocol - REDCap Variable Definitions

Codebook for `heartland_data_dictionary.csv` (v1.0.2)

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Overview

This codebook documents every field in the HEARTLAND REDCap instrument template. For each field you will find: REDCap variable name, form, field type, units/range, clinical definition, and the HEARTLAND Protocol v3.3 reference.

Companion file: `instruments/heartland_data_dictionary.csv` - the machine-readable data dictionary that produces this codebook.

Cutoffs and weighted points for the HEARTLAND Risk Score reproduce **Table 1 of Protocol v3.3** (*Cureus* 2026; DOI 10.7759/cureus.104817; current Zenodo deposit 10.5281/zenodo.19101219).

Form 1 - Enrollment (`enrollment`)

Variable	Type	Range	Definition	Ref
<code>record_id</code>	auto	integer	REDCap record identifier, auto-assigned on import	-
<code>enr_date</code>	date	YYYY-MM-DD	Date patient was enrolled in the HEARTLAND cohort	Module 1

Variable	Type	Range	Definition	Ref
consent_date	date	YYYY-MM-DD	Date of signed informed consent	Module 1
facility_name	text	free text (PHI if site is identifiable)	Enrolling facility	Module 2
facility_cah	yesno	Y / N	Designated Critical Access Hospital (CMS CAH program)	Module 2
facility_tier	radio	1 / 2 / 3	Implementation Protocol tier (Tier 1 Minimal, Tier 2 Standard, Tier 3 Advanced)	Table 2
state	text	2-letter US state	Facility US state code	Module 2
county_fips	text	5-digit	County FIPS code for geographic analysis	Module 2
rural_urban	radio	1-4	USDA Rural-Urban Commuting Area category collapsed to Metro / Micro / Small-town / Rural	Module 2
age	integer	18-110 years	Age in completed years at enrollment	Demographics

Variable	Type	Range	Definition	Ref
sex	radio	Male / Female / Other	Sex at birth as documented in the medical record	Demographics
race	checkbox	NIH categories	Race - check all that apply (NIH categories)	Demographics
ethnicity	radio	Hispanic / Not / Unknown	Ethnicity per NIH categories	Demographics

Form 2 - Baseline Assessment (baseline)

This form captures the **10 HEARTLAND Risk Variables** (Protocol v3.3 Table 1). The risk score is **0-18 points weighted** (not a simple binary sum). Three tiers drive monitoring intensity: Low 0-4, Moderate 5-8, High ≥ 9 .

Risk variables (10)

Variable	Rule $\rightarrow$ Points
age (from enrollment)	+2 if ≥ 75
bl_prior_hf_hosp_6mo	+3 if prior HF hospitalization within last 6 months
bl_egfr	+2 if < 45 mL/min/1.73 m ²
bl_np_type + bl_np_value	+2 if BNP ≥ 500 OR NT-proBNP ≥ 1500 pg/mL
bl_sbp_admit	+2 if admission systolic BP < 100 mmHg
bl_diabetes	+1 if diabetes mellitus
bl_lvef_pct	+2 if LVEF $< 30\%$
bl_ckm_stage	+2 if AHA CKM Stage 3 or 4
bl_distance_cardio_mi	+1 if distance to nearest cardiologist > 50 miles
bl_social_support_limited	+1 if lives alone OR limited social support

Calculated fields

- **bl_risk_score** (calc, 0-18): weighted sum using the REDCap formula embedded in the data dictionary.
- **bl_risk_tier** (text + @CALCTEXT): "low" if <5, "moderate" if 5-8, "high" if >=9.

Supporting baseline fields

Variable	Definition
bl_lvef_category	HFrEF (<=40%) / HFmrEF (41-49%) / HFpEF (>=50%) - drives GDMT branching
bl_enrichd_score	Optional 7-item ENRICH Social Support Scale total (range 7-35); <=19 suggests limited support (Mitchell et al., 2003)
bl_pro_consent	Gate for KCCQ-12 fields on monthly follow-up and outcomes forms

Form 3 - GDMT Status (**gdmt_status**)

Captures current heart-failure guideline-directed medical therapy. For HFrEF, the four foundational classes (Protocol Module 4): RAS inhibitor (preferably ARNI), beta-blocker, MRA, SGLT2 inhibitor. HFpEF-specific fields branch on **bl_lvef_category** = HFpEF.

RAS inhibitor (**gdmt_arni_acei_arb_***)

Drug (ARNI / ACEi / ARB), total daily dose (mg), site-specified target (protocol default sacubitril/valsartan 97/103 BID = 200 mg/day), start date. Fields suppressed when drug = None.

Beta-blocker (**gdmt_bb_***)

Carvedilol, metoprolol succinate, or bisoprolol. Protocol target: carvedilol 25 mg BID (50 mg BID if >85 kg).

MRA (**gdmt_mra_***)

Spironolactone, eplerenone, or finerenone. Finerenone indicated for HFpEF with CKD (FINEARTS-HF). Target dose spironolactone 25-50 mg/day.

SGLT2 inhibitor (gdmt_sglit2_*)

Dapagliflozin 10 mg or empagliflozin 10 mg (no titration). Safety gate: eGFR >20.

HFpEF-specific

- gdmt_hfpef_glp1: GLP-1 RA (e.g., semaglutide) - STEP-HFpEF indication if BMI >=30 (HFpEF only).

Summary

- gdmt_classes_count (calc, 0-4): number of the four HFpEF classes currently prescribed (non-zero drug selected).
- gdmt_generic_bridge: Y if patient is on the \$4 generic bridge (ACE-I + BB + spironolactone) while pursuing optimal agents.
- gdmt_init_tier: site's initiation strategy - Tier 1 (>=2 classes before discharge, prioritize SGLT2i + BB), Tier 2 (all tolerated, uptitrate 2-4 wk), Tier 3 (rapid-sequence per STRONG-HF).

Form 4 - Monthly Follow-up (monthly_followup, repeating instrument, months 1-12)

Encounter metadata

Variable	Definition
mo_event_number	1-12 - month number
mo_date	Visit / contact date
mo_track	A (digital: Bluetooth + app) vs B (analog: paper diary + phone). Assigned at enrollment; can switch.

Vitals (all tracks)

mo_weight_lb (50-600), mo_sbp (60-250), mo_dbp (30-150), mo_hr (30-200), mo_spo2 (50-100).

Labs (optional - "if available")

mo_egfr, mo_k (K+ mEq/L), mo_bnp - quarterly if available. Labs are not required for protocol adherence.

GDMT & events

- `mo_gdmt_change` (Y/N): any change this month (titration, discontinuation, new agent).
- `mo_gdmt_change_notes`: free text, gated by `mo_gdmt_change = 1`.
- `mo_red_flag_count`: count of red-flag events.
- `mo_red_flag_types` (checkbox, gated if count > 0): weight gain, orthopnea, PND, edema, syncope, angina, arrhythmia, other.
- `mo_hosp_any`, `mo_hosp_hf` (gated on `any=Y`), `mo_ed_any`, `mo_ed_hf` (gated on `any=Y`).

Patient-reported outcomes (consent-gated)

`mo_kccq12_score` (0-100): site-computed KCCQ-12 summary score. Shown only when `bl_pro_consent = Yes`. The template does not reproduce KCCQ-12 questions or scoring instructions; sites must obtain and administer any required licensed instrument separately.

Form 5 - Outcomes at 12 Months (`outcomes_12mo`)

Primary endpoint

- `out_vital_status`: Alive / Deceased / Lost to follow-up.

Secondary endpoints

Variable	Definition
<code>out_death_date</code>	Date of death (if deceased)
<code>out_death_cardiovascular</code>	Cardiovascular cause of death (Y/N)
<code>out_hf_hosp_count</code>	HF hospitalizations in 12 months
<code>out_hf_ed_count</code>	HF-related ED visits in 12 months
<code>out_days_alive_oh</code>	Days alive and out of hospital (0-365)
<code>out_gdmt_optimized</code>	Y if ≥3 HFrEF classes at target dose at 12 months
<code>out_kccq12_12mo</code>	KCCQ-12 at 12 months (consent-gated)

Branching logic summary

Condition	Fields gated
bl_lvef_category = "HFpEF"	gdm_t_hfpef_glp1
bl_pro_consent = 1	mo_kccq12_score, out_kccq12_12mo
mo_gdm_t_change = 1	mo_gdm_t_change_notes
mo_red_flag_count > 0	mo_red_flag_types
mo_hosp_any = 1	mo_hosp_hf
mo_ed_any = 1	mo_ed_hf
out_vital_status = 2	out_death_date, out_death_cardiovascular
Drug selector != 0 (None)	Associated dose / target / start-date fields per class

Calculated fields summary

Variable	Formula source
bl_risk_score	Sum of 10 weighted indicator rules (Protocol Table 1)
bl_risk_tier	@CALCTEXT over bl_risk_score: <5 low / 5-8 moderate / >=9 high
gdm_t_classes_count	Count of non-None drug selections across RAS / BB / MRA / SGLT2

Data-integrity notes

- Field names follow REDCap conventions (lowercase, underscores, <=26 chars).
- All eight full-date fields and **facility_name** are marked as identifiers in the REDCap data dictionary; local sites remain responsible for their own HIPAA/IRB privacy configuration.
- The **record_id** column is REDCap's mandatory primary key.
- Repeating instrument (**monthly_followup**) must be enabled in the target project's Project Setup after import.
- Sample data in **examples/sample_data.csv** is fully synthetic (no real patients).

References

- Ferreira VM. *HEARTLAND Protocol: A Tiered Clinical Implementation Toolkit for Primary Care-Led Heart Failure Management in Ru-*

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- Mitchell PH, Powell L, Blumenthal J, et al. *A short social support measure for patients recovering from myocardial infarction: the ENRICH Social Support Inventory*. J Cardiopulm Rehabil. 2003;23(6):398-403.
- Solomon SD, et al. *Finerenone in HFpEF* (FINEARTS-HF). N Engl J Med. 2024.
- Mebazaa A, et al. *Safety, tolerability and efficacy of up-titration of guideline-directed medical therapies for acute heart failure* (STRONG-HF). Lancet. 2022;400:1938-52.